TUTORIAL

Molecular dynamics basics

> Updated for EasyHybrid 3.2.0 (2026-10-26). The figures and numbers come from a run of this tutorial with the 3.2.0 development version.

This tutorial runs a short molecular dynamics (MD) simulation of a peptide in water and shows the three steps that come before almost any analysis of a trajectory:

1. **reimaging**: put the molecules that crossed the periodic box back into it, with the peptide at the center;
2. **alignment**: superpose every frame on a reference, removing the rotation and translation of the peptide;
3. **RMSD**: how far the structure moves from the reference during the run.

The system is `examples/tutorials/04_md_basics/1UAO_OPLS_folded_solvated.pkl` in the EasyHybrid folder: chignolin (PDB 1UAO, 10 residues) in a cubic box of 665 TIP3P-like water molecules and 2 Na^+, OPLS-AA, 2135 atoms, periodic boundary conditions. The MD takes about 2.5 minutes on one core.

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1. Open the system

File → Open and choose the `.pkl` file. The status bar shows `force field: OPLS`, `PBC: True` and `symmetry: Cubic`.

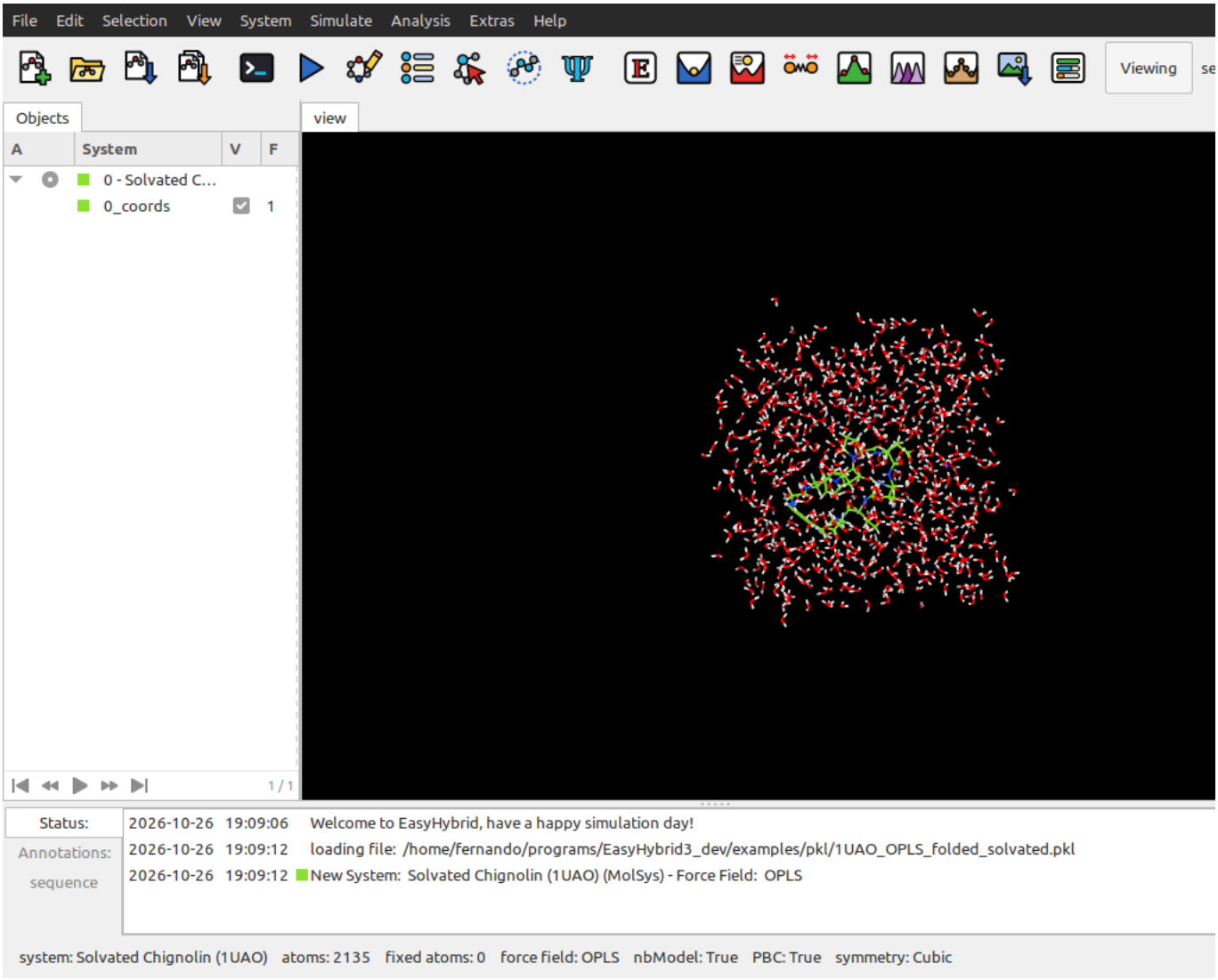


Figure 1. Chignolin in its water box.

2. Run the dynamics

Simulate → Molecular Dynamics (Figure 2):

Field	Value	Why
Integrator	Langevin Dynamics	keeps the temperature at 300 K
Number of Steps	10000	10 ps with a 1 fs step
Time Step (ps)	0.001	pDynamo has no SHAKE: 1 fs
Temperature	300 K	
Collision Frequency	25 ps^{-1}	Langevin friction
Log Frequency	500	one line of the log every 0.5 ps
Save Trajectory	on, name `1UAO_md`, format pDynamo / pkl	
Trajectory Frequency	100	one frame every 0.1 ps: 101 frames

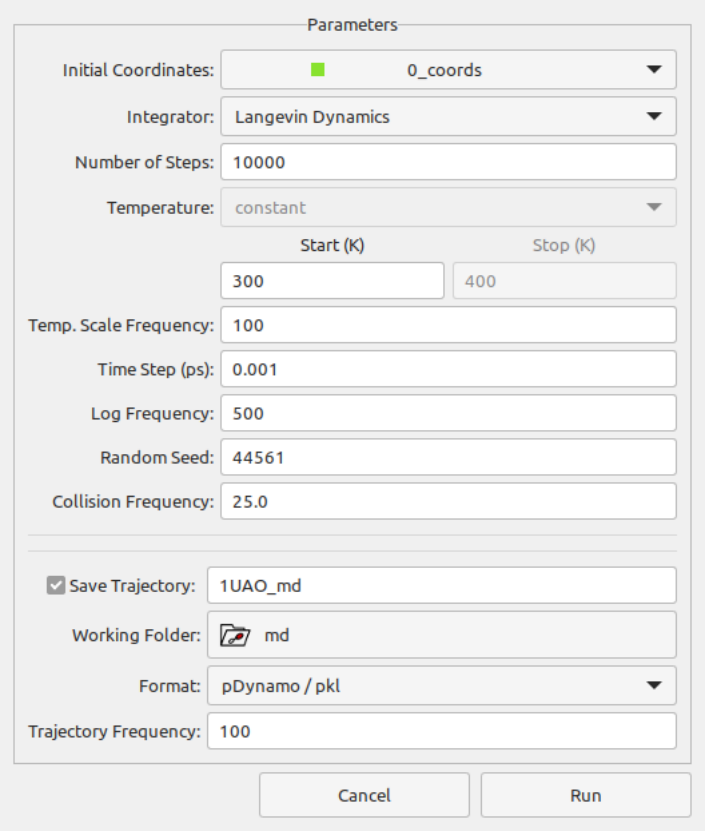


Figure 2. The Molecular Dynamics window.

Click **Run**. The job runs in the background (see the Process Manager). The log gives the temperature every 0.5 ps; it fluctuates around 295 K (285–306 K) in this run.

When the job ends, click **Import...** in the *Finished!* dialog and **Import**: a new object with 101 frames is added. Play it with the buttons under the object list.

> **10 ps is a demonstration, not a simulation.** A real study first equilibrates the system (heating, then a constant-pressure run) and then collects nanoseconds of production.

3. Reimaging

Go to the last frame. Some water molecules have left the box and the box looks frayed (Figure 3): during the MD each molecule moves freely and the periodic images are only used to compute the interactions.

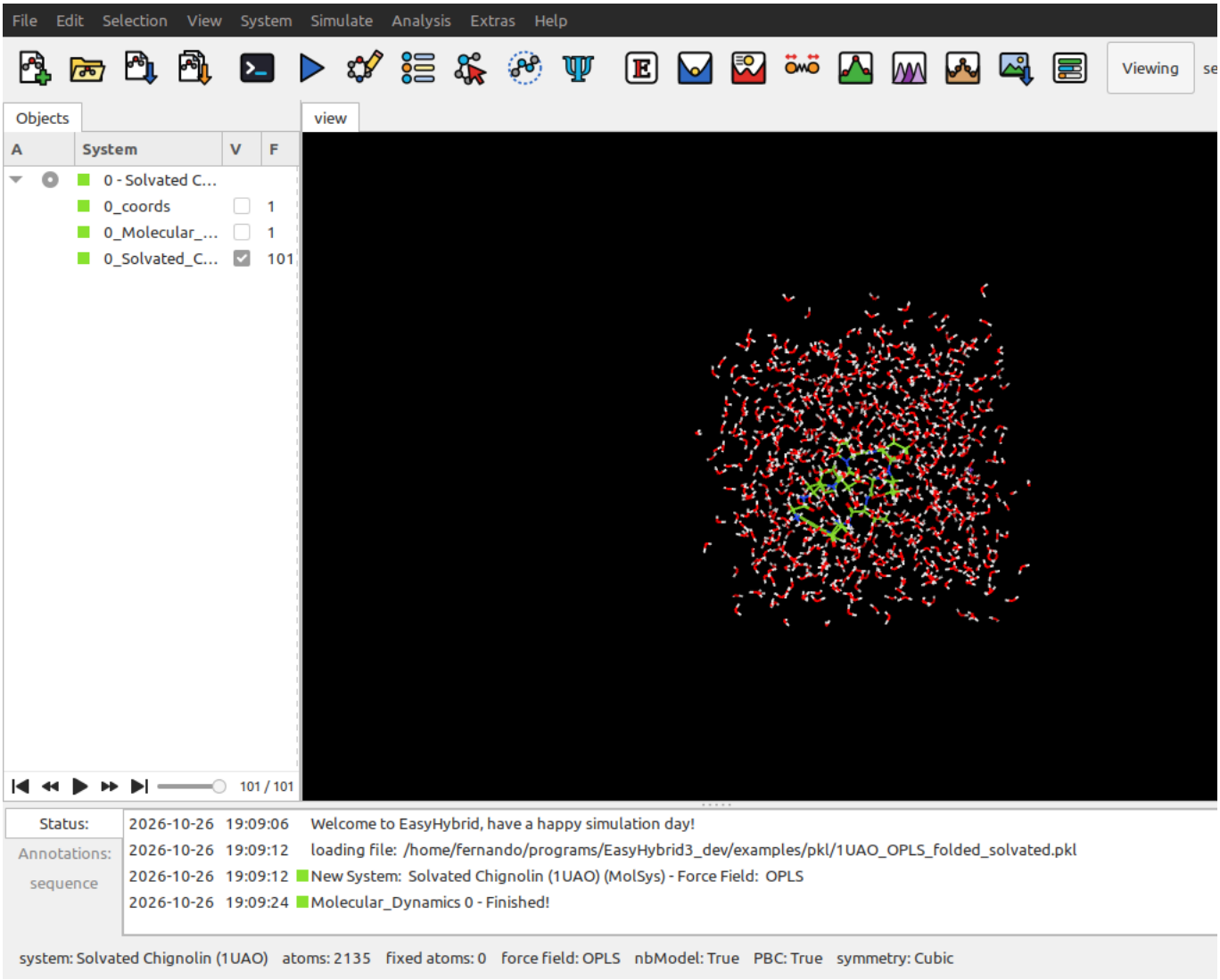


Figure 3. The last frame as written by the MD.

1. Select the peptide. In the EasyHybrid Terminal (**View → EasyHybrid Terminal**): `select obj=2 not_resn=HOH,NA,CL,SOD,CLA` (138 atoms; `obj=2` is the trajectory object, see `list`).
2. **Analysis → Reimaging (Wrapping)**: the **Coordinates** is the trajectory; tick **Center the selection in the box** (Figure 4), and click **Reimaging**.
3. Type `deselect` in the terminal.

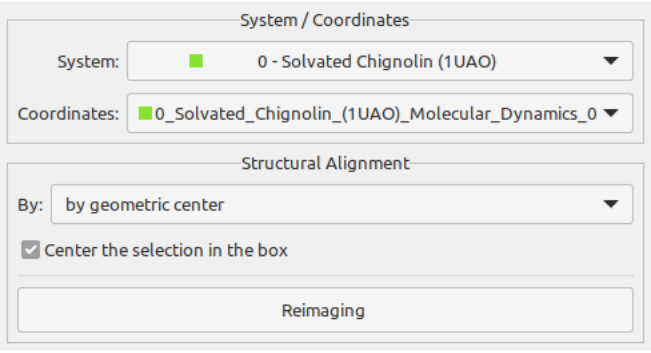


Figure 4. Reimaging with the peptide at the center of the box.

In every frame the peptide is moved to the center of the box and each molecule is put back inside it by its geometric center (Figure 5).

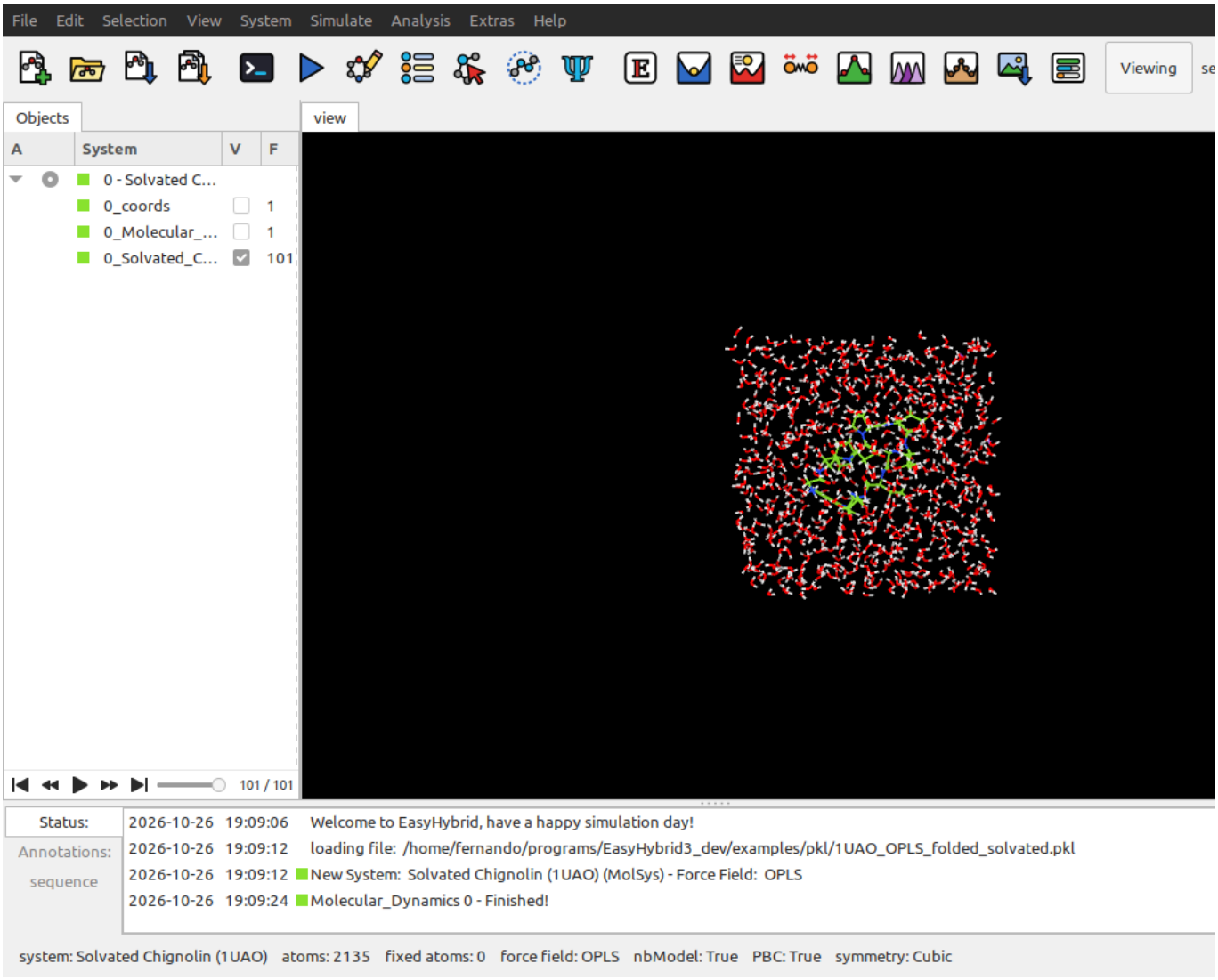


Figure 5. The same frame after reimaging.

4. RMSD without alignment

Select the Cα atoms (`select obj=2 name=CA` , 10 atoms) and open **Analysis → RMSD tool**. Keep the reference **frame 0**, untick **Superpose each frame on the reference (Kabsch)** and click **Run** (Figure 6).

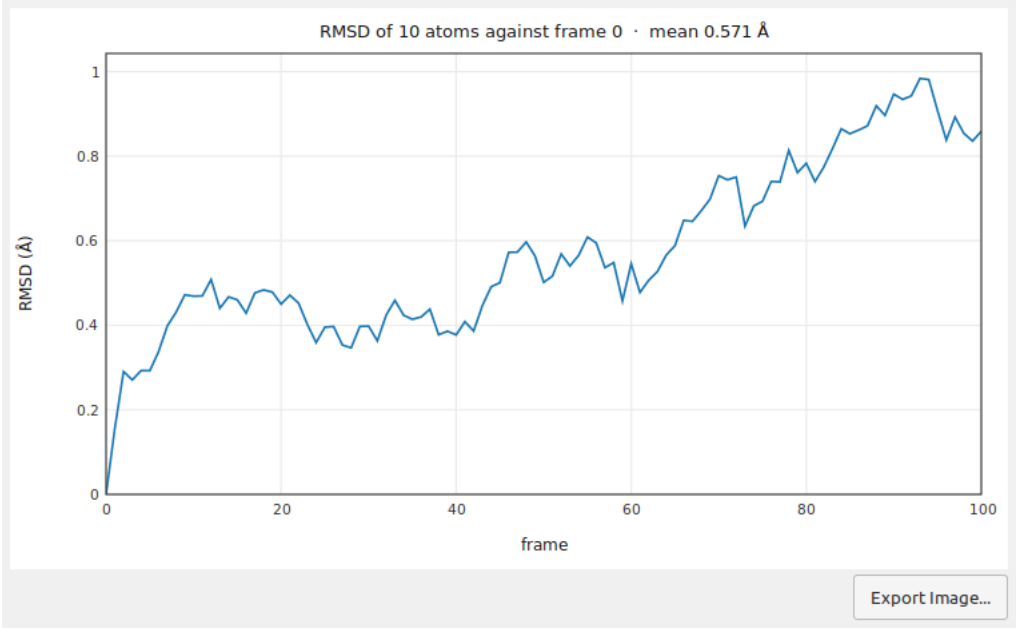


Figure 6. RMSD of the Cα atoms against frame 0, without superposition: mean 0.57 Å.

The RMSD grows steadily to almost 1 Å. Most of it is not a change of shape: the peptide is centered in each frame by reimaging, but it still **rotates** in the box, and a rotation moves every atom.

5. Align the trajectory

Analysis → Align Trajectory: **By** *C-alpha only*, **Reference** frame 0, **Align** (Figure 7). Every frame is superposed on frame 0 by least squares on the Cα atoms; the frames of the object are replaced by the aligned ones.

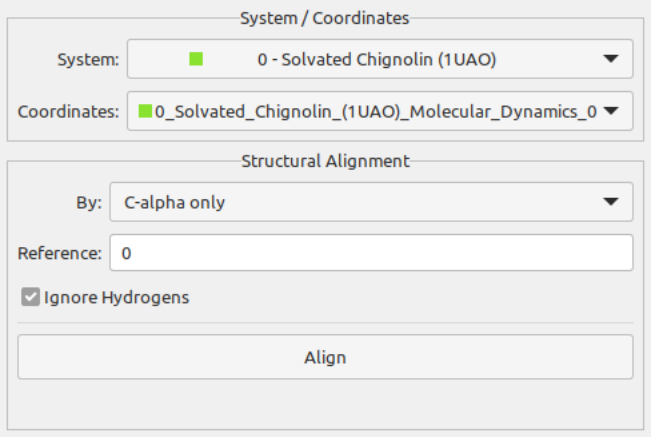


Figure 7. Align Trajectory on the Cα atoms.

Run the RMSD again, with the same options (Figure 8).

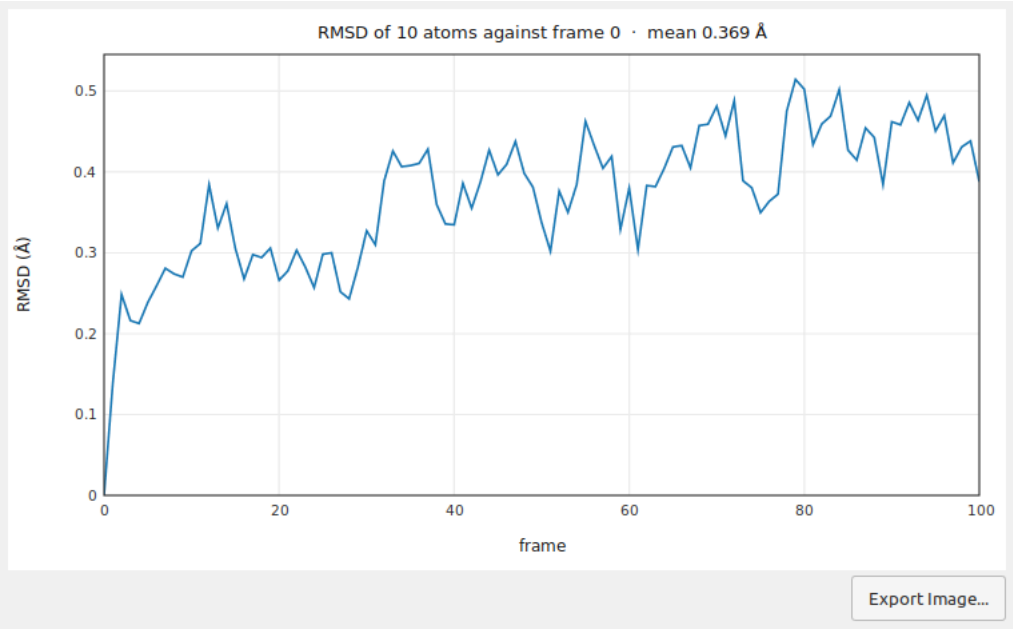


Figure 8. RMSD after the alignment: mean 0.37 Å, maximum 0.52 Å.

The RMSD now measures only the change of shape: the hairpin stays within 0.5 Å of its starting structure, as expected for a stable fold over 10 ps.

	Mean RMSD (Å)	Maximum (Å)
No superposition	0.57	0.98
After Align Trajectory	0.37	0.52
Kabsch superposition in the RMSD tool	0.37	0.52

The option **Superpose each frame on the reference (Kabsch)** of the RMSD tool, on by default, does the same superposition frame by frame without changing the object: use it for a quick RMSD, and **Align Trajectory** before analyses that compare coordinates frame by frame (RMSF, average structure, PCA).

Notes

- **What to align on.** Align on the atoms whose motion you want to remove: the backbone or the Cα of the protein, never the water. For a protein–ligand complex, align on the protein and measure the RMSD of the ligand without superposition: it shows how far the ligand moved in the site.
- **The Trajectory Analysis window** (**Analysis → Trajectory Analysis…**) gathers 15 more analyses: radius of gyration, secondary structure, hydrogen bonds, contacts, clustering, PCA, B-factors and more.
- The tutorial Radial distribution functions uses a trajectory of this same system.

EasyHybrid 3.2.0 · built 2026-10-29 from `dev` @ `3539da4` · https://github.com/ferbachega/EasyHybrid3